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ФЕДЕРАЛЬНОЕ ГОСУДАРСТВЕННОЕ АВТОНОМНОЕ ОБРАЗОВАТЕЛЬНОЕ
УЧРЕЖДЕНИЕ ВЫСШЕГО ОБРАЗОВАНИЯ
НАЦИОНАЛЬНЫЙ ИССЛЕДОВАТЕЛЬСКИЙ УНИВЕРСИТЕТ ИТМО**

ОТЧЕТ

ПО ЛАБОРАТОРНОЙ РАБОТЕ № 3

«Проектирование и реализация баз данных в СУБД PostgreSQL»

по дисциплине «Проектирование и реализация баз данных»

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Группа К3241

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Образовательная программа Мобильные и сетевые технологии

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Лабораторная работа №3.1

Цель работы: овладеть практическими навыками установки СУБД PostgreSQL и создания базы данных в pgadmin 4.

Практическое задание:

1. Установить СУБД PostgreSQL 1X.
2. Создать базу данных с использованием pgadmin 4

Описание предметной области: Сеть автомастерских осуществляет ремонт автомобилей, используя для этих целей штат мастеров и свои мастерские. Стоимость ремонта включает цену деталей и стоимость работы.

Заработная плата мастеров составляет 50% стоимости работы.

С клиентом заключается договор на выполнение авторемонтных и профилактических работ, который сопровождается администратором. В каждом договоре может быть несколько видов услуг. Для выполнения видов работ могут требоваться детали или расходные материалы, которые предоставляет либо клиент, либо автомастерская. Если детали предоставляет автомастерская, то их стоимость включается в смету по договору.

Каждый вид работ могут выполнять разные мастера, в зависимости от их специализации. Распределение мастеров выполняет администратор.

БД должна содержать следующий минимальный набор сведений:
Табельный номер сотрудника. ФИО сотрудника. Должность. Разряд мастера. Специализация. Адрес автомастерской. Дата заказа. Гос. Номер автомобиля. Марка. Мощность автомобиля. Год выпуска. Цвет автомобиля. Дата принятия в ремонт. Плановая дата окончания ремонта. Фактическая дата окончания ремонта. Вид ремонта. Стоимость вида ремонта. Название детали. Цена детали. Марка и модель автомобиля. Страна производителя. Госномер автомобиля. ФИО владельца. Номер телефона владельца. E-mail владельца. Должность сотрудника. Количество ставок (по штатному расписанию).

Ход выполнения работы

1. СУБД PostgreSQL была установлена с использованием пакетного менеджера Homebrew с помощью команды:
`brew install postgresql`
2. После установки был запущен сервер PostgreSQL:
`brew services start postgresql`
3. Далее был выполнен запуск графической среды pgAdmin 4.
4. В pgAdmin было выполнено подключение к локальному серверу PostgreSQL с использованием пользователя postgres.
5. В дереве объектов pgAdmin был выбран узел Databases, после чего через контекстное меню была выполнена команда Create → Database
6. В открывшемся окне была создана база данных:

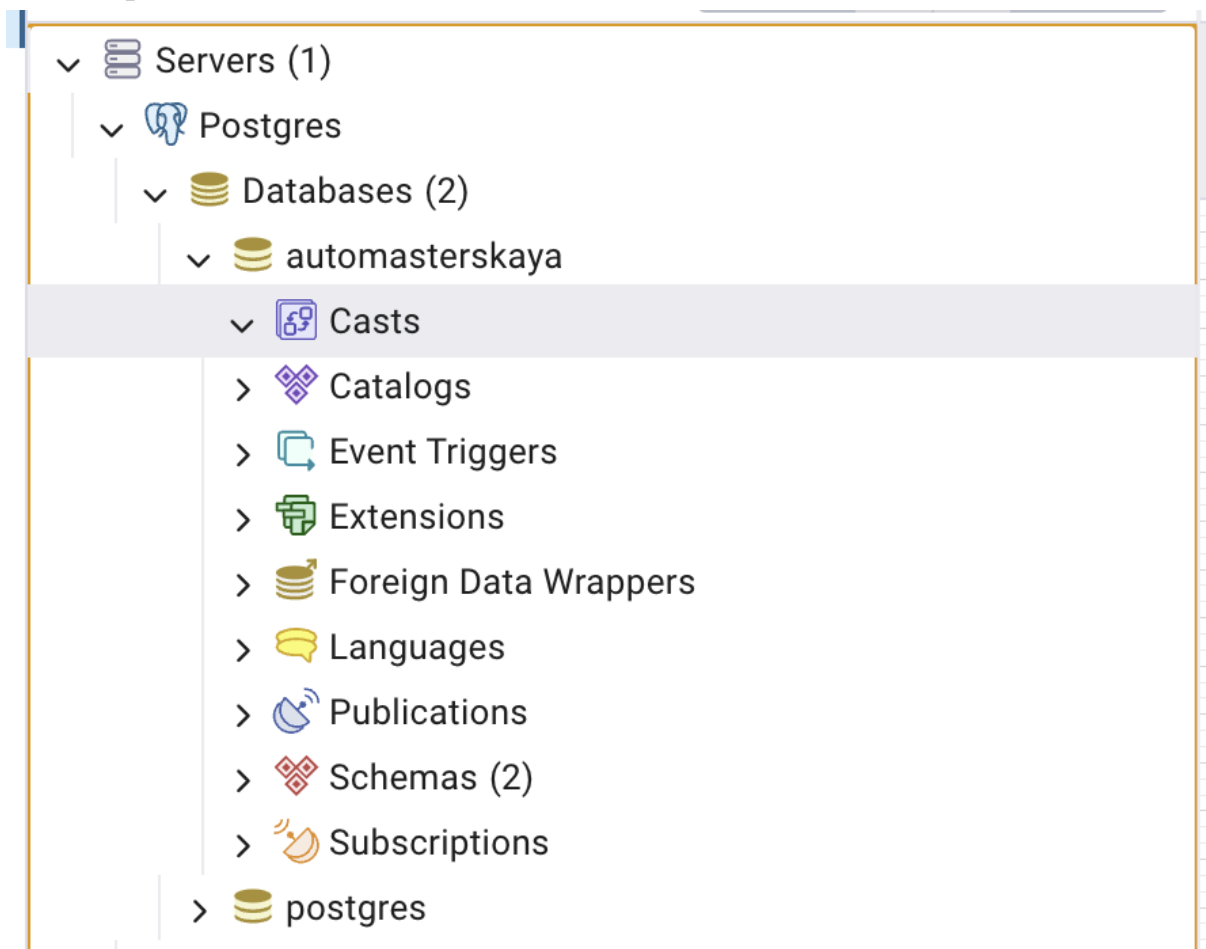


Рис 1 - Обзор объектов pgAdmin с созданной базой данных

Лабораторная работа №3.2

Цель работы: овладеть практическими навыками создания таблиц базы данных PostgreSQL 1X, заполнения их рабочими данными, резервного копирования и восстановления

Практическое задание:

1. Создать базу данных с использованием pgAdmin 4 (согласно индивидуальному заданию).
2. Создать схему в составе базы данных.
3. Создать таблицы базы данных.
4. Установить ограничения на данные: Primary Key, Unique, Check, Foreign Key.
5. Заполнить таблицы БД рабочими данными.
6. Создать резервную копию БД.

Указание:

Создать две резервные копии:

- с расширением CUSTOM для восстановления БД;
- с расширением для листинга (в отчете);
- при создании резервных копий БД настроить параметры Dump options для Type of objects и Queries .

7. Восстановить БД.

Ход выполнения работы

1. Реализация модели

На основе разработанной инфологической модели предметной области «Сеть автомастерских» была выполнена физическая реализация базы данных в СУБД PostgreSQL с использованием pgAdmin 4.

При реализации модели составные ключи, использованные на этапе проектирования, были заменены на простые суррогатные ключи с

автоинкрементом (SERIAL). Это позволило упростить структуру таблиц и корректно реализовать связи между сущностями.

Для логической организации объектов базы данных была создана отдельная схема, в которой размещены все таблицы.

2. Нормализация

Выбрана нормализация 3NF, так как такое представление позволяет уменьшить избыточность данных и обеспечить их целостность.

3. ERD-диаграмма

На основе реализованной структуры базы данных средствами pgAdmin была сгенерирована ERD-диаграмма, отражающая таблицы и связи между ними.

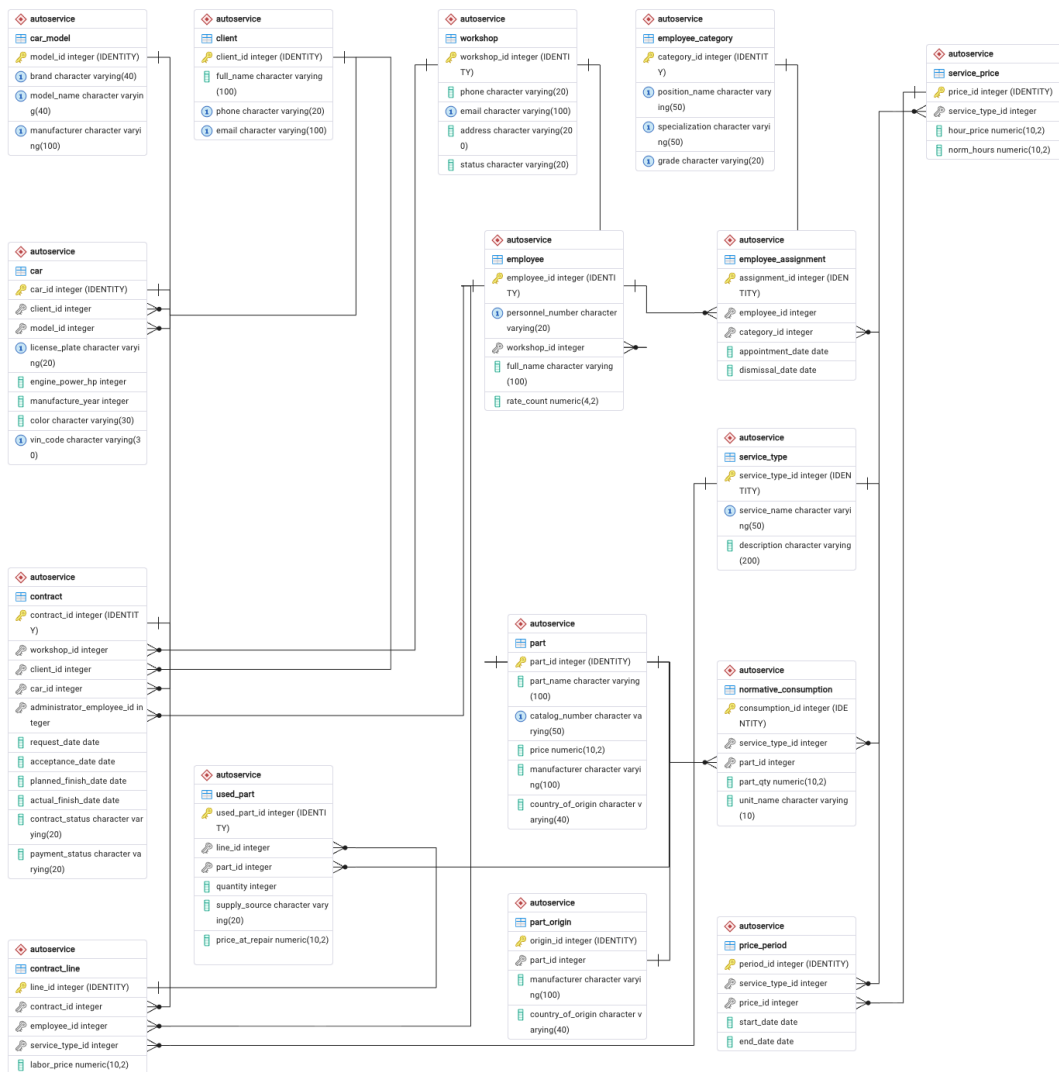


Рис 2 – ERD диаграмма

4. В рамках реализации базы данных были созданы таблицы, соответствующие сущностям предметной области:

- car - автомобили;
- car_model - модели автомобилей;
- client - владельцы автомобилей;
- contract - договоры на выполнение ремонтных работ;
- contract_line - виды работ в рамках договора;
- employee - сотрудники;

- employee_assignment - назначение сотрудников на выполнение работ;
- employee_category - категории и квалификация сотрудников;
- normative_consumption - нормативы расхода деталей и материалов;
- part - детали и расходные материалы;
- part_origin - источник предоставления деталей (клиент или мастерская);
- price_period - периоды действия цен;
- service_price - стоимость услуг;
- service_type - виды ремонтных работ;
- used_part - использование деталей в ремонте;
- workshop - автомастерские.

5. Ограничения

Для обеспечения целостности данных были заданы следующие ограничения:

PRIMARY KEY

Реализованы во всех таблицах в виде уникальных идентификаторов.

FOREIGN KEY

Обеспечивают связи между таблицами, в частности:

- заказ связан с автомобилем, владельцем и администратором;
- автомобиль связан с владельцем;
- работы связаны с заказами;
- детали связаны с работами;
- сотрудники связаны с должностями и специализациями.

UNIQUE

Использованы для предотвращения дублирования:

- государственный номер автомобиля;
- контактные данные владельца.

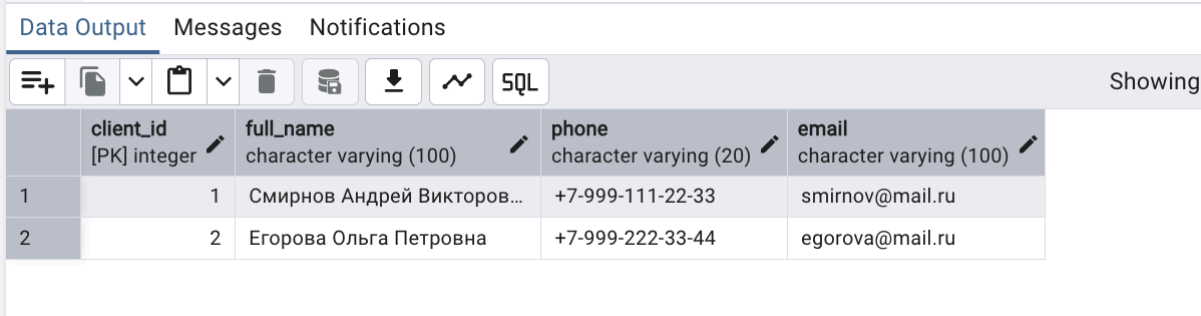
CHECK

Обеспечивают корректность данных:

- положительные значения стоимости;
- корректные временные интервалы ремонта;
- допустимые значения атрибутов.

6. Заполнение данными

Заполнение таблиц выполнялось с использованием SQL-запросов INSERT через Query Tool в pgAdmin.



	client_id [PK] integer	full_name character varying (100)	phone character varying (20)	email character varying (100)
1	1	Смирнов Андрей Викторов...	+7-999-111-22-33	smirnov@mail.ru
2	2	Егорова Ольга Петровна	+7-999-222-33-44	egorova@mail.ru

Рис 3 – Вывод команды SELECT * from autoservice.client

7. Резервное копирование и восстановление

Были созданы две резервные копии базы данных:

- в формате Custom - для восстановления;
- в формате Plain - для анализа структуры и содержимого.

Резервное копирование выполнено с помощью инструмента Backup в pgAdmin.

Выводы по ЛР №3.1, №3.2

В ходе выполнения лабораторных работ №3.1 и №3.2 были получены практические навыки работы с СУБД PostgreSQL и средой pgAdmin 4: выполнена установка системы с использованием Homebrew, произведён запуск сервера и создана база данных, а также реализована база данных на основе инфологической модели предметной области «Сеть

автоматических», включающая создание таблиц, установление связей между ними и задание ограничений целостности (PRIMARY KEY, FOREIGN KEY, UNIQUE, CHECK); дополнительно было выполнено заполнение таблиц тестовыми данными с использованием SQL-запросов, проведена проверка корректности данных, а также освоены методы резервного копирования базы данных и её последующего восстановления, что позволяет применять полученные знания для проектирования и реализации реляционных баз данных.

```
--
-- PostgreSQL database dump
--

\restrict
dNNF5sppDcAKdQIL5xQxdSepiaRk4rAMP5u00u7RmRL2SAcXnNzpaPze5x
KWcrQ

-- Dumped from database version 15.17 (Homebrew)
-- Dumped by pg_dump version 18.2

-- Started on 2026-04-13 17:19:07 MSK

SET statement_timeout = 0;
SET lock_timeout = 0;
SET idle_in_transaction_session_timeout = 0;
SET transaction_timeout = 0;
SET client_encoding = 'UTF8';
SET standard_conforming_strings = on;
SELECT pg_catalog.set_config('search_path', '', false);
SET check_function_bodies = false;
SET xmloption = content;
SET client_min_messages = warning;
SET row_security = off;

--
-- TOC entry 6 (class 2615 OID 16389)
-- Name: autoservice; Type: SCHEMA; Schema: -; Owner: bareckijmaksim
--

CREATE SCHEMA autoservice;

ALTER SCHEMA autoservice OWNER TO bareckijmaksim;

SET default_tablespace = '';
```

```
SET default_table_access_method = heap;
```

```
--  
-- TOC entry 228 (class 1259 OID 16458)  
-- Name: car; Type: TABLE; Schema: autoservice; Owner: bareckijmaksim  
--
```

```
CREATE TABLE autoservice.car (  
    car_id integer NOT NULL,  
    client_id integer NOT NULL,  
    model_id integer NOT NULL,  
    license_plate character varying(20) NOT NULL,  
    engine_power_hp integer NOT NULL,  
    manufacture_year integer NOT NULL,  
    color character varying(30) NOT NULL,  
    vin_code character varying(30),  
    CONSTRAINT chk_car_power CHECK ((engine_power_hp > 0)),  
    CONSTRAINT chk_car_year CHECK (((manufacture_year >= 1950) AND  
(manufacture_year <= 2100)))  
);
```

```
ALTER TABLE autoservice.car OWNER TO bareckijmaksim;
```

```
--  
-- TOC entry 227 (class 1259 OID 16457)  
-- Name: car_car_id_seq; Type: SEQUENCE; Schema: autoservice; Owner:  
bareckijmaksim  
--
```

```
ALTER TABLE autoservice.car ALTER COLUMN car_id ADD GENERATED  
ALWAYS AS IDENTITY (  
    SEQUENCE NAME autoservice.car_car_id_seq  
    START WITH 1  
    INCREMENT BY 1  
    NO MINVALUE  
    NO MAXVALUE
```

```

    CACHE 1
);

--
-- TOC entry 226 (class 1259 OID 16450)
-- Name: car_model; Type: TABLE; Schema: autoservice; Owner:
bareckijmaksim
--

CREATE TABLE autoservice.car_model (
    model_id integer NOT NULL,
    brand character varying(40) NOT NULL,
    model_name character varying(40) NOT NULL,
    manufacturer character varying(100) NOT NULL
);

ALTER TABLE autoservice.car_model OWNER TO bareckijmaksim;

--
-- TOC entry 225 (class 1259 OID 16449)
-- Name: car_model_model_id_seq; Type: SEQUENCE; Schema: autoservice;
Owner: bareckijmaksim
--

ALTER TABLE autoservice.car_model ALTER COLUMN model_id ADD
GENERATED ALWAYS AS IDENTITY (
    SEQUENCE NAME autoservice.car_model_model_id_seq
    START WITH 1
    INCREMENT BY 1
    NO MINVALUE
    NO MAXVALUE
    CACHE 1
);

```

```
--  
-- TOC entry 224 (class 1259 OID 16440)  
-- Name: client; Type: TABLE; Schema: autoservice; Owner: bareckijmaksim  
--
```

```
CREATE TABLE autoservice.client (  
    client_id integer NOT NULL,  
    full_name character varying(100) NOT NULL,  
    phone character varying(20) NOT NULL,  
    email character varying(100)  
);
```

```
ALTER TABLE autoservice.client OWNER TO bareckijmaksim;
```

```
--  
-- TOC entry 223 (class 1259 OID 16439)  
-- Name: client_client_id_seq; Type: SEQUENCE; Schema: autoservice;  
Owner: bareckijmaksim  
--
```

```
ALTER TABLE autoservice.client ALTER COLUMN client_id ADD  
GENERATED ALWAYS AS IDENTITY (  
    SEQUENCE NAME autoservice.client_client_id_seq  
    START WITH 1  
    INCREMENT BY 1  
    NO MINVALUE  
    NO MAXVALUE  
    CACHE 1  
);
```

```
--  
-- TOC entry 230 (class 1259 OID 16480)  
-- Name: contract; Type: TABLE; Schema: autoservice; Owner: bareckijmaksim  
--
```

```

CREATE TABLE autoservice.contract (
    contract_id integer NOT NULL,
    workshop_id integer NOT NULL,
    client_id integer NOT NULL,
    car_id integer NOT NULL,
    administrator_employee_id integer NOT NULL,
    request_date date NOT NULL,
    acceptance_date date NOT NULL,
    planned_finish_date date,
    actual_finish_date date,
    contract_status character varying(20) NOT NULL,
    payment_status character varying(20) NOT NULL,
    CONSTRAINT chk_contract_dates CHECK (((acceptance_date >=
request_date) AND ((planned_finish_date IS NULL) OR (planned_finish_date
>= acceptance_date)) AND ((actual_finish_date IS NULL) OR
(actual_finish_date >= acceptance_date)))),
    CONSTRAINT chk_contract_status CHECK (((contract_status)::text = ANY
((ARRAY['новый'::character varying, 'в работе'::character varying,
'завершён'::character varying, 'отменён'::character varying]))::text[])),
    CONSTRAINT chk_payment_status CHECK (((payment_status)::text =
ANY ((ARRAY['не оплачено'::character varying, 'частично'::character
varying, 'оплачено'::character varying]))::text[])))
);

```

```

ALTER TABLE autoservice.contract OWNER TO bareckijmaksim;

```

```

--
-- TOC entry 229 (class 1259 OID 16479)
-- Name: contract_contract_id_seq; Type: SEQUENCE; Schema: autoservice;
Owner: bareckijmaksim
--

```

```

ALTER TABLE autoservice.contract ALTER COLUMN contract_id ADD
GENERATED ALWAYS AS IDENTITY (
    SEQUENCE NAME autoservice.contract_contract_id_seq
    START WITH 1

```

```
INCREMENT BY 1
NO MINVALUE
NO MAXVALUE
CACHE 1
);
```

```
--
-- TOC entry 238 (class 1259 OID 16547)
-- Name: contract_line; Type: TABLE; Schema: autoservice; Owner:
bareckijmaksim
--
```

```
CREATE TABLE autoservice.contract_line (
    line_id integer NOT NULL,
    contract_id integer NOT NULL,
    employee_id integer NOT NULL,
    service_type_id integer NOT NULL,
    labor_price numeric(10,2) NOT NULL,
    CONSTRAINT chk_contract_line_price CHECK ((labor_price >=
(0)::numeric))
);
```

```
ALTER TABLE autoservice.contract_line OWNER TO bareckijmaksim;
```

```
--
-- TOC entry 237 (class 1259 OID 16546)
-- Name: contract_line_line_id_seq; Type: SEQUENCE; Schema: autoservice;
Owner: bareckijmaksim
--
```

```
ALTER TABLE autoservice.contract_line ALTER COLUMN line_id ADD
GENERATED ALWAYS AS IDENTITY (
    SEQUENCE NAME autoservice.contract_line_line_id_seq
    START WITH 1
    INCREMENT BY 1
```

```

NO MINVALUE
NO MAXVALUE
CACHE 1
);

--
-- TOC entry 220 (class 1259 OID 16408)
-- Name: employee; Type: TABLE; Schema: autoservice; Owner:
bareckijmaksim
--

CREATE TABLE autoservice.employee (
    employee_id integer NOT NULL,
    personnel_number character varying(20) NOT NULL,
    workshop_id integer NOT NULL,
    full_name character varying(100) NOT NULL,
    rate_count numeric(4,2) DEFAULT 1.00 NOT NULL,
    CONSTRAINT chk_employee_rate CHECK (((rate_count > (0)::numeric)
AND (rate_count <= (3)::numeric)))
);

ALTER TABLE autoservice.employee OWNER TO bareckijmaksim;

--
-- TOC entry 222 (class 1259 OID 16423)
-- Name: employee_assignment; Type: TABLE; Schema: autoservice; Owner:
bareckijmaksim
--

CREATE TABLE autoservice.employee_assignment (
    assignment_id integer NOT NULL,
    employee_id integer NOT NULL,
    category_id integer NOT NULL,
    appointment_date date NOT NULL,
    dismissal_date date,

```



```
CONSTRAINT chk_assignment_dates CHECK (((dismissal_date IS NULL)
OR (dismissal_date >= appointment_date)))
);
```

```
ALTER TABLE autoservice.employee_assignment OWNER TO
bareckijmaksim;
```

```
--
-- TOC entry 221 (class 1259 OID 16422)
-- Name: employee_assignment_assignment_id_seq; Type: SEQUENCE;
Schema: autoservice; Owner: bareckijmaksim
--
```

```
ALTER TABLE autoservice.employee_assignment ALTER COLUMN
assignment_id ADD GENERATED ALWAYS AS IDENTITY (
    SEQUENCE NAME autoservice.employee_assignment_assignment_id_seq
    START WITH 1
    INCREMENT BY 1
    NO MINVALUE
    NO MAXVALUE
    CACHE 1
);
```

```
--
-- TOC entry 218 (class 1259 OID 16400)
-- Name: employee_category; Type: TABLE; Schema: autoservice; Owner:
bareckijmaksim
--
```

```
CREATE TABLE autoservice.employee_category (
    category_id integer NOT NULL,
    position_name character varying(50) NOT NULL,
    specialization character varying(50),
    grade character varying(20)
);
```

```
ALTER TABLE autoservice.employee_category OWNER TO bareckijmaksim;
```

```
--
```

```
-- TOC entry 217 (class 1259 OID 16399)
```

```
-- Name: employee_category_category_id_seq; Type: SEQUENCE; Schema:  
autoservice; Owner: bareckijmaksim
```

```
--
```

```
ALTER TABLE autoservice.employee_category ALTER COLUMN  
category_id ADD GENERATED ALWAYS AS IDENTITY (  
    SEQUENCE NAME autoservice.employee_category_category_id_seq  
    START WITH 1  
    INCREMENT BY 1  
    NO MINVALUE  
    NO MAXVALUE  
    CACHE 1  
);
```

```
--
```

```
-- TOC entry 219 (class 1259 OID 16407)
```

```
-- Name: employee_employee_id_seq; Type: SEQUENCE; Schema:  
autoservice; Owner: bareckijmaksim
```

```
--
```

```
ALTER TABLE autoservice.employee ALTER COLUMN employee_id ADD  
GENERATED ALWAYS AS IDENTITY (  
    SEQUENCE NAME autoservice.employee_employee_id_seq  
    START WITH 1  
    INCREMENT BY 1  
    NO MINVALUE  
    NO MAXVALUE  
    CACHE 1  
);
```

```
--  
-- TOC entry 244 (class 1259 OID 16591)  
-- Name: normative_consumption; Type: TABLE; Schema: autoservice; Owner:  
bareckijmaksim  
--
```

```
CREATE TABLE autoservice.normative_consumption (  
    consumption_id integer NOT NULL,  
    service_type_id integer NOT NULL,  
    part_id integer NOT NULL,  
    part_qty numeric(10,2) NOT NULL,  
    unit_name character varying(10) NOT NULL,  
    CONSTRAINT chk_normative_qty CHECK ((part_qty > (0)::numeric))  
);
```

```
ALTER TABLE autoservice.normative_consumption OWNER TO  
bareckijmaksim;
```

```
--  
-- TOC entry 243 (class 1259 OID 16590)  
-- Name: normative_consumption_consumption_id_seq; Type: SEQUENCE;  
Schema: autoservice; Owner: bareckijmaksim  
--
```

```
ALTER TABLE autoservice.normative_consumption ALTER COLUMN  
consumption_id ADD GENERATED ALWAYS AS IDENTITY (  
    SEQUENCE NAME  
autoservice.normative_consumption_consumption_id_seq  
    START WITH 1  
    INCREMENT BY 1  
    NO MINVALUE  
    NO MAXVALUE  
    CACHE 1  
);
```

```
--  
-- TOC entry 240 (class 1259 OID 16569)  
-- Name: part; Type: TABLE; Schema: autoservice; Owner: bareckijmaksim  
--
```

```
CREATE TABLE autoservice.part (  
    part_id integer NOT NULL,  
    part_name character varying(100) NOT NULL,  
    catalog_number character varying(50) NOT NULL,  
    price numeric(10,2) NOT NULL,  
    manufacturer character varying(100) NOT NULL,  
    country_of_origin character varying(40) NOT NULL,  
    CONSTRAINT chk_part_price CHECK ((price >= (0)::numeric))  
);
```

```
ALTER TABLE autoservice.part OWNER TO bareckijmaksim;
```

```
--  
-- TOC entry 242 (class 1259 OID 16578)  
-- Name: part_origin; Type: TABLE; Schema: autoservice; Owner:  
bareckijmaksim  
--
```

```
CREATE TABLE autoservice.part_origin (  
    origin_id integer NOT NULL,  
    part_id integer NOT NULL,  
    manufacturer character varying(100) NOT NULL,  
    country_of_origin character varying(40) NOT NULL  
);
```

```
ALTER TABLE autoservice.part_origin OWNER TO bareckijmaksim;
```

```
--  
-- TOC entry 241 (class 1259 OID 16577)
```

-- Name: part_origin_origin_id_seq; Type: SEQUENCE; Schema: autoservice;
Owner: bareckijmaksim

--

```
ALTER TABLE autoservice.part_origin ALTER COLUMN origin_id ADD  
GENERATED ALWAYS AS IDENTITY (  
    SEQUENCE NAME autoservice.part_origin_origin_id_seq  
    START WITH 1  
    INCREMENT BY 1  
    NO MINVALUE  
    NO MAXVALUE  
    CACHE 1  
);
```

--

-- TOC entry 239 (class 1259 OID 16568)
-- Name: part_part_id_seq; Type: SEQUENCE; Schema: autoservice; Owner:
bareckijmaksim

--

```
ALTER TABLE autoservice.part ALTER COLUMN part_id ADD  
GENERATED ALWAYS AS IDENTITY (  
    SEQUENCE NAME autoservice.part_part_id_seq  
    START WITH 1  
    INCREMENT BY 1  
    NO MINVALUE  
    NO MAXVALUE  
    CACHE 1  
);
```

--

-- TOC entry 236 (class 1259 OID 16530)
-- Name: price_period; Type: TABLE; Schema: autoservice; Owner:
bareckijmaksim

--

```
CREATE TABLE autoservice.price_period (  
    period_id integer NOT NULL,  
    service_type_id integer NOT NULL,  
    price_id integer NOT NULL,  
    start_date date NOT NULL,  
    end_date date,  
    CONSTRAINT chk_price_period_dates CHECK (((end_date IS NULL) OR  
(end_date >= start_date)))  
);
```

```
ALTER TABLE autoservice.price_period OWNER TO bareckijmaksim;
```

```
--  
-- TOC entry 235 (class 1259 OID 16529)  
-- Name: price_period_period_id_seq; Type: SEQUENCE; Schema:  
autoservice; Owner: bareckijmaksim  
--
```

```
ALTER TABLE autoservice.price_period ALTER COLUMN period_id ADD  
GENERATED ALWAYS AS IDENTITY (  
    SEQUENCE NAME autoservice.price_period_period_id_seq  
    START WITH 1  
    INCREMENT BY 1  
    NO MINVALUE  
    NO MAXVALUE  
    CACHE 1  
);
```

```
--  
-- TOC entry 234 (class 1259 OID 16517)  
-- Name: service_price; Type: TABLE; Schema: autoservice; Owner:  
bareckijmaksim  
--
```

```
CREATE TABLE autoservice.service_price (  
    price_id integer NOT NULL,  
    service_type_id integer NOT NULL,  
    hour_price numeric(10,2) NOT NULL,  
    norm_hours numeric(10,2) NOT NULL,  
    CONSTRAINT chk_service_price_hour CHECK ((hour_price >  
(0)::numeric)),  
    CONSTRAINT chk_service_price_norm CHECK ((norm_hours >  
(0)::numeric))  
);
```

```
ALTER TABLE autoservice.service_price OWNER TO bareckijmaksim;
```

```
--  
-- TOC entry 233 (class 1259 OID 16516)  
-- Name: service_price_price_id_seq; Type: SEQUENCE; Schema: autoservice;  
Owner: bareckijmaksim  
--
```

```
ALTER TABLE autoservice.service_price ALTER COLUMN price_id ADD  
GENERATED ALWAYS AS IDENTITY (  
    SEQUENCE NAME autoservice.service_price_price_id_seq  
    START WITH 1  
    INCREMENT BY 1  
    NO MINVALUE  
    NO MAXVALUE  
    CACHE 1  
);
```

```
--  
-- TOC entry 232 (class 1259 OID 16509)  
-- Name: service_type; Type: TABLE; Schema: autoservice; Owner:  
bareckijmaksim  
--
```

```
CREATE TABLE autoservice.service_type (  
    service_type_id integer NOT NULL,  
    service_name character varying(50) NOT NULL,  
    description character varying(200)  
);
```

```
ALTER TABLE autoservice.service_type OWNER TO bareckijmaksim;
```

```
--  
-- TOC entry 231 (class 1259 OID 16508)  
-- Name: service_type_service_type_id_seq; Type: SEQUENCE; Schema:  
autoservice; Owner: bareckijmaksim  
--
```

```
ALTER TABLE autoservice.service_type ALTER COLUMN service_type_id  
ADD GENERATED ALWAYS AS IDENTITY (  
    SEQUENCE NAME autoservice.service_type_service_type_id_seq  
    START WITH 1  
    INCREMENT BY 1  
    NO MINVALUE  
    NO MAXVALUE  
    CACHE 1  
);
```

```
--  
-- TOC entry 246 (class 1259 OID 16610)  
-- Name: used_part; Type: TABLE; Schema: autoservice; Owner:  
bareckijmaksim  
--
```

```
CREATE TABLE autoservice.used_part (  
    used_part_id integer NOT NULL,  
    line_id integer NOT NULL,  
    part_id integer NOT NULL,  
    quantity integer NOT NULL,
```



```
supply_source character varying(20) NOT NULL,  
price_at_repair numeric(10,2) NOT NULL,  
CONSTRAINT chk_used_part_price CHECK ((price_at_repair >=  
(0)::numeric)),  
CONSTRAINT chk_used_part_qty CHECK ((quantity > 0)),  
CONSTRAINT chk_used_part_source CHECK (((supply_source)::text =  
ANY ((ARRAY['клиент'::character varying, 'автомастерская'::character  
varying]))::text[])))  
);
```

```
ALTER TABLE autoservice.used_part OWNER TO bareckijmaksim;
```

```
--  
-- TOC entry 245 (class 1259 OID 16609)  
-- Name: used_part_used_part_id_seq; Type: SEQUENCE; Schema:  
autoservice; Owner: bareckijmaksim  
--
```

```
ALTER TABLE autoservice.used_part ALTER COLUMN used_part_id ADD  
GENERATED ALWAYS AS IDENTITY (  
    SEQUENCE NAME autoservice.used_part_used_part_id_seq  
    START WITH 1  
    INCREMENT BY 1  
    NO MINVALUE  
    NO MAXVALUE  
    CACHE 1  
);
```

```
--  
-- TOC entry 247 (class 1259 OID 16628)  
-- Name: vw_employee_salary; Type: VIEW; Schema: autoservice; Owner:  
bareckijmaksim  
--
```

```
CREATE VIEW autoservice.vw_employee_salary AS
```

```

SELECT cl.line_id,
       cl.contract_id,
       cl.employee_id,
       e.full_name,
       cl.labor_price,
       round((cl.labor_price * 0.50), 2) AS salary_amount
FROM (autoservice.contract_line cl
      JOIN autoservice.employee e ON ((e.employee_id = cl.employee_id)));

```

```

ALTER VIEW autoservice.vw_employee_salary OWNER TO bareckijmaksim;

```

```

--
-- TOC entry 248 (class 1259 OID 24576)
-- Name: vw_mechanic_favorite_car_model; Type: VIEW; Schema:
autoservice; Owner: bareckijmaksim
--

```

```

CREATE VIEW autoservice.vw_mechanic_favorite_car_model AS
WITH mechanic_model_stats AS (
    SELECT e.employee_id,
           e.full_name AS mechanic_name,
           cm.brand,
           cm.model_name,
           count(*) AS repair_count,
           row_number() OVER (PARTITION BY e.employee_id ORDER BY
(count(*)) DESC, cm.brand, cm.model_name) AS rn
    FROM (((((autoservice.contract_line cl
              JOIN autoservice.contract c ON ((c.contract_id = cl.contract_id)))
              JOIN autoservice.car ca ON ((ca.car_id = c.car_id)))
              JOIN autoservice.car_model cm ON ((cm.model_id = ca.model_id)))
              JOIN autoservice.employee e ON ((e.employee_id = cl.employee_id)))
              JOIN autoservice.employee_assignment ea ON ((ea.employee_id =
e.employee_id))))
              JOIN autoservice.employee_category ec ON ((ec.category_id =
ea.category_id)))
    WHERE ((ec.position_name)::text = 'Мастер'::text)

```

```

        GROUP BY e.employee_id, e.full_name, cm.brand, cm.model_name
    )
SELECT mechanic_model_stats.mechanic_name,
       mechanic_model_stats.brand,
       mechanic_model_stats.model_name,
       mechanic_model_stats.repair_count
FROM mechanic_model_stats
WHERE (mechanic_model_stats.rn = 1);

```

```

ALTER VIEW autoservice.vw_mechanic_favorite_car_model OWNER TO
bareckijmaksim;

```

```

--
-- TOC entry 249 (class 1259 OID 24581)
-- Name: vw_mechanic_monthly_bonus; Type: VIEW; Schema: autoservice;
Owner: bareckijmaksim
--

```

```

CREATE VIEW autoservice.vw_mechanic_monthly_bonus AS
WITH monthly_jobs AS (
    SELECT e.employee_id,
           e.full_name,
           cl.labor_price,
           c.planned_finish_date,
           c.actual_finish_date
    FROM (((autoservice.employee e
           JOIN autoservice.contract_line cl ON ((cl.employee_id =
e.employee_id)))
           JOIN autoservice.contract c ON ((c.contract_id = cl.contract_id)))
           JOIN autoservice.employee_assignment ea ON ((ea.employee_id =
e.employee_id)))
           JOIN autoservice.employee_category ec ON ((ec.category_id =
ea.category_id)))
    WHERE (((ec.position_name)::text = 'Мастер'::text) AND
(c.actual_finish_date >= '2025-03-01'::date) AND (c.actual_finish_date <
'2025-04-01'::date))

```

```

), salary_calc AS (
  SELECT monthly_jobs.employee_id,
         monthly_jobs.full_name,
         round(sum((monthly_jobs.labor_price * 0.50)), 2) AS salary_amount,
         bool_and((monthly_jobs.actual_finish_date <=
monthly_jobs.planned_finish_date)) AS all_in_time
  FROM monthly_jobs
 WHERE ((monthly_jobs.actual_finish_date IS NOT NULL) AND
(monthly_jobs.planned_finish_date IS NOT NULL))
  GROUP BY monthly_jobs.employee_id, monthly_jobs.full_name
)
SELECT salary_calc.employee_id,
       salary_calc.full_name,
       salary_calc.salary_amount,
       CASE
         WHEN salary_calc.all_in_time THEN
round((salary_calc.salary_amount * 0.10), 2)
         ELSE 0.00
       END AS bonus_amount
  FROM salary_calc;

```

```

ALTER VIEW autoservice.vw_mechanic_monthly_bonus OWNER TO
bareckijmaksim;

```

```

--
-- TOC entry 216 (class 1259 OID 16391)
-- Name: workshop; Type: TABLE; Schema: autoservice; Owner:
bareckijmaksim
--

```

```

CREATE TABLE autoservice.workshop (
  workshop_id integer NOT NULL,
  phone character varying(20) NOT NULL,
  email character varying(100),
  address character varying(200) NOT NULL,
  status character varying(20) NOT NULL,

```

```
CONSTRAINT chk_workshop_status CHECK (((status)::text = ANY
((ARRAY['действует'::character varying, 'в ремонте'::character varying,
'закрыта'::character varying])::text[])))
);
```

```
ALTER TABLE autoservice.workshop OWNER TO bareckijmaksim;
```

```
--
-- TOC entry 215 (class 1259 OID 16390)
-- Name: workshop_workshop_id_seq; Type: SEQUENCE; Schema:
autoservice; Owner: bareckijmaksim
--
```

```
ALTER TABLE autoservice.workshop ALTER COLUMN workshop_id ADD
GENERATED ALWAYS AS IDENTITY (
    SEQUENCE NAME autoservice.workshop_workshop_id_seq
    START WITH 1
    INCREMENT BY 1
    NO MINVALUE
    NO MAXVALUE
    CACHE 1
);
```

```
--
-- TOC entry 4011 (class 0 OID 16458)
-- Dependencies: 228
-- Data for Name: car; Type: TABLE DATA; Schema: autoservice; Owner:
bareckijmaksim
--
```

```
COPY autoservice.car (car_id, client_id, model_id, license_plate,
engine_power_hp, manufacture_year, color, vin_code) FROM stdin;
1 1 1 A111AA77 181 2018 черный VIN0000000000000001
2 2 2 B222BB77 123 2020 белый VIN0000000000000002
\.
```

--
-- TOC entry 4009 (class 0 OID 16450)
-- Dependencies: 226
-- Data for Name: car_model; Type: TABLE DATA; Schema: autoservice;
Owner: bareckijmaksim
--

COPY autoservice.car_model (model_id, brand, model_name, manufacturer)
FROM stdin;

1 Toyota Camry Toyota Motor Corporation
2 Kia Rio Kia Corporation
\.

--
-- TOC entry 4007 (class 0 OID 16440)
-- Dependencies: 224
-- Data for Name: client; Type: TABLE DATA; Schema: autoservice; Owner:
bareckijmaksim
--

COPY autoservice.client (client_id, full_name, phone, email) FROM stdin;
1 Смирнов Андрей Викторович +7-999-111-22-33 smirnov@mail.ru
2 Егорова Ольга Петровна +7-999-222-33-44 egorova@mail.ru
\.

--
-- TOC entry 4013 (class 0 OID 16480)
-- Dependencies: 230
-- Data for Name: contract; Type: TABLE DATA; Schema: autoservice; Owner:
bareckijmaksim
--

```
COPY autoservice.contract (contract_id, workshop_id, client_id, car_id,
administrator_employee_id, request_date, acceptance_date,
planned_finish_date, actual_finish_date, contract_status, payment_status)
FROM stdin;
```

```
1 1 1 1 1 2025-03-01 2025-03-02 2025-03-03 2025-03-03
завершён оплачено
3 1 1 1 1 2025-03-10 2025-03-11 2025-03-12 2025-03-12
завершён оплачено
2 1 2 2 1 2025-03-05 2025-03-06 2025-03-08 2025-03-08
отменён частично
\.
```

--

-- TOC entry 4021 (class 0 OID 16547)

-- Dependencies: 238

-- Data for Name: contract_line; Type: TABLE DATA; Schema: autoservice;
Owner: bareckijmaksim

--

```
COPY autoservice.contract_line (line_id, contract_id, employee_id,
service_type_id, labor_price) FROM stdin;
```

```
1 1 2 1 1800.00
2 2 3 2 7700.00
3 2 4 3 1800.00
4 3 2 1 1800.00
5 2 4 3 1800.00
\.
```

--

-- TOC entry 4003 (class 0 OID 16408)

-- Dependencies: 220

-- Data for Name: employee; Type: TABLE DATA; Schema: autoservice;
Owner: bareckijmaksim

--

COPY autoservice.employee (employee_id, personnel_number, workshop_id, full_name, rate_count) FROM stdin;

1	T-001	1	Иванова Марина Сергеевна	1.00
2	T-002	1	Петров Алексей Николаевич	1.00
3	T-003	1	Сидоров Максим Олегович	1.50
4	T-004	2	Кузнецов Илья Павлович	1.00

\.

--

-- TOC entry 4005 (class 0 OID 16423)

-- Dependencies: 222

-- Data for Name: employee_assignment; Type: TABLE DATA; Schema: autoservice; Owner: bareckijmaksim

--

COPY autoservice.employee_assignment (assignment_id, employee_id, category_id, appointment_date, dismissal_date) FROM stdin;

1	1	1	2024-01-10	\N
2	2	2	2024-01-12	\N
3	3	3	2024-02-01	\N
4	4	4	2024-02-10	\N

\.

--

-- TOC entry 4001 (class 0 OID 16400)

-- Dependencies: 218

-- Data for Name: employee_category; Type: TABLE DATA; Schema: autoservice; Owner: bareckijmaksim

--

COPY autoservice.employee_category (category_id, position_name, specialization, grade) FROM stdin;

1	Администратор	Оформление договоров	\N
2	Мастер	Двигатель	5
3	Мастер	Ходовая часть	4

4 Мастер Диагностика 5

\.

--

-- TOC entry 4027 (class 0 OID 16591)

-- Dependencies: 244

-- Data for Name: normative_consumption; Type: TABLE DATA; Schema: autoservice; Owner: bareckijmaksim

--

COPY autoservice.normative_consumption (consumption_id, service_type_id, part_id, part_qty, unit_name) FROM stdin;

1 1 1 1.00 шт

2 1 2 1.00 шт

3 2 3 2.00 шт

\.

--

-- TOC entry 4023 (class 0 OID 16569)

-- Dependencies: 240

-- Data for Name: part; Type: TABLE DATA; Schema: autoservice; Owner: bareckijmaksim

--

COPY autoservice.part (part_id, part_name, catalog_number, price, manufacturer, country_of_origin) FROM stdin;

1 Масляный фильтр OF-1001 650.00 Mann Германия

2 Моторное масло 5W-30 OIL-5W30 3200.00 Shell Нидерланды

3 Стойка стабилизатора ST-2002 2400.00 Lemforder Германия

\.

--

-- TOC entry 4025 (class 0 OID 16578)

-- Dependencies: 242

-- Data for Name: part_origin; Type: TABLE DATA; Schema: autoservice;
Owner: bareckijmaksim

--

COPY autoservice.part_origin (origin_id, part_id, manufacturer,
country_of_origin) FROM stdin;

1	1	Mann	Германия
2	2	Shell	Нидерланды
3	3	Lemforder	Германия

\.

--

-- TOC entry 4019 (class 0 OID 16530)

-- Dependencies: 236

-- Data for Name: price_period; Type: TABLE DATA; Schema: autoservice;
Owner: bareckijmaksim

--

COPY autoservice.price_period (period_id, service_type_id, price_id,
start_date, end_date) FROM stdin;

1	1	1	2025-01-01	\N
2	2	2	2025-01-01	\N
3	3	3	2025-01-01	\N

\.

--

-- TOC entry 4017 (class 0 OID 16517)

-- Dependencies: 234

-- Data for Name: service_price; Type: TABLE DATA; Schema: autoservice;
Owner: bareckijmaksim

--

COPY autoservice.service_price (price_id, service_type_id, hour_price,
norm_hours) FROM stdin;

1	1	1500.00	1.20
---	---	---------	------

2 2 2200.00 3.50

3 3 1800.00 1.00

\\.

--

-- TOC entry 4015 (class 0 OID 16509)

-- Dependencies: 232

-- Data for Name: service_type; Type: TABLE DATA; Schema: autoservice;
Owner: bareckijmaksim

--

COPY autoservice.service_type (service_type_id, service_name, description)
FROM stdin;

1 Замена масла Плановая замена моторного масла и фильтра

2 Ремонт подвески Ремонт элементов подвески

3 Компьютерная диагностика Проверка электронных систем
автомобиля

\\.

--

-- TOC entry 4029 (class 0 OID 16610)

-- Dependencies: 246

-- Data for Name: used_part; Type: TABLE DATA; Schema: autoservice;
Owner: bareckijmaksim

--

COPY autoservice.used_part (used_part_id, line_id, part_id, quantity,
supply_source, price_at_repair) FROM stdin;

1 1 1 1 автомастерская 650.00

2 1 2 1 автомастерская 3200.00

\\.

--

-- TOC entry 3999 (class 0 OID 16391)

```
-- Dependencies: 216
-- Data for Name: workshop; Type: TABLE DATA; Schema: autoservice;
Owner: bareckijmaksim
--

COPY autoservice.workshop (workshop_id, phone, email, address, status)
FROM stdin;
1 +7-495-111-11-11 central@autoservice.ru г. Москва, ул. Ленина, 10
действует
2 +7-495-222-22-22 south@autoservice.ru г. Москва, ул. Южная, 15
действует
\.
```

```
--
-- TOC entry 4035 (class 0 OID 0)
-- Dependencies: 227
-- Name: car_car_id_seq; Type: SEQUENCE SET; Schema: autoservice;
Owner: bareckijmaksim
--
```

```
SELECT pg_catalog.setval('autoservice.car_car_id_seq', 2, true);
```

```
--
-- TOC entry 4036 (class 0 OID 0)
-- Dependencies: 225
-- Name: car_model_model_id_seq; Type: SEQUENCE SET; Schema:
autoservice; Owner: bareckijmaksim
--
```

```
SELECT pg_catalog.setval('autoservice.car_model_model_id_seq', 2, true);
```

```
--
-- TOC entry 4037 (class 0 OID 0)
-- Dependencies: 223
```

-- Name: client_client_id_seq; Type: SEQUENCE SET; Schema: autoservice;
Owner: bareckijmaksim

--

SELECT pg_catalog.setval('autoservice.client_client_id_seq', 2, true);

--

-- TOC entry 4038 (class 0 OID 0)

-- Dependencies: 229

-- Name: contract_contract_id_seq; Type: SEQUENCE SET; Schema:
autoservice; Owner: bareckijmaksim

--

SELECT pg_catalog.setval('autoservice.contract_contract_id_seq', 3, true);

--

-- TOC entry 4039 (class 0 OID 0)

-- Dependencies: 237

-- Name: contract_line_line_id_seq; Type: SEQUENCE SET; Schema:
autoservice; Owner: bareckijmaksim

--

SELECT pg_catalog.setval('autoservice.contract_line_line_id_seq', 5, true);

--

-- TOC entry 4040 (class 0 OID 0)

-- Dependencies: 221

-- Name: employee_assignment_assignment_id_seq; Type: SEQUENCE SET;
Schema: autoservice; Owner: bareckijmaksim

--

SELECT

pg_catalog.setval('autoservice.employee_assignment_assignment_id_seq', 4,
true);

```
--  
-- TOC entry 4041 (class 0 OID 0)  
-- Dependencies: 217  
-- Name: employee_category_category_id_seq; Type: SEQUENCE SET;  
Schema: autoservice; Owner: bareckijmaksim  
--  
  
SELECT pg_catalog.setval('autoservice.employee_category_category_id_seq',  
4, true);
```

```
--  
-- TOC entry 4042 (class 0 OID 0)  
-- Dependencies: 219  
-- Name: employee_employee_id_seq; Type: SEQUENCE SET; Schema:  
autoservice; Owner: bareckijmaksim  
--
```

```
SELECT pg_catalog.setval('autoservice.employee_employee_id_seq', 4, true);
```

```
--  
-- TOC entry 4043 (class 0 OID 0)  
-- Dependencies: 243  
-- Name: normative_consumption_consumption_id_seq; Type: SEQUENCE  
SET; Schema: autoservice; Owner: bareckijmaksim  
--
```

```
SELECT  
pg_catalog.setval('autoservice.normative_consumption_consumption_id_seq', 3,  
true);
```

```
--  
-- TOC entry 4044 (class 0 OID 0)
```

```

-- Dependencies: 241
-- Name: part_origin_origin_id_seq; Type: SEQUENCE SET; Schema:
autoservice; Owner: bareckijmaksim
--

SELECT pg_catalog.setval('autoservice.part_origin_origin_id_seq', 3, true);

--
-- TOC entry 4045 (class 0 OID 0)
-- Dependencies: 239
-- Name: part_part_id_seq; Type: SEQUENCE SET; Schema: autoservice;
Owner: bareckijmaksim
--

SELECT pg_catalog.setval('autoservice.part_part_id_seq', 3, true);

--
-- TOC entry 4046 (class 0 OID 0)
-- Dependencies: 235
-- Name: price_period_period_id_seq; Type: SEQUENCE SET; Schema:
autoservice; Owner: bareckijmaksim
--

SELECT pg_catalog.setval('autoservice.price_period_period_id_seq', 3, true);

--
-- TOC entry 4047 (class 0 OID 0)
-- Dependencies: 233
-- Name: service_price_price_id_seq; Type: SEQUENCE SET; Schema:
autoservice; Owner: bareckijmaksim
--

SELECT pg_catalog.setval('autoservice.service_price_price_id_seq', 3, true);

```

```
--  
-- TOC entry 4048 (class 0 OID 0)  
-- Dependencies: 231  
-- Name: service_type_service_type_id_seq; Type: SEQUENCE SET; Schema:  
autoservice; Owner: bareckijmaksim  
--
```

```
SELECT pg_catalog.setval('autoservice.service_type_service_type_id_seq', 3,  
true);
```

```
--  
-- TOC entry 4049 (class 0 OID 0)  
-- Dependencies: 245  
-- Name: used_part_used_part_id_seq; Type: SEQUENCE SET; Schema:  
autoservice; Owner: bareckijmaksim  
--
```

```
SELECT pg_catalog.setval('autoservice.used_part_used_part_id_seq', 3, true);
```

```
--  
-- TOC entry 4050 (class 0 OID 0)  
-- Dependencies: 215  
-- Name: workshop_workshop_id_seq; Type: SEQUENCE SET; Schema:  
autoservice; Owner: bareckijmaksim  
--
```

```
SELECT pg_catalog.setval('autoservice.workshop_workshop_id_seq', 2, true);
```

```
--  
-- TOC entry 3800 (class 2606 OID 16466)  
-- Name: car_car_license_plate_key; Type: CONSTRAINT; Schema:  
autoservice; Owner: bareckijmaksim  
--
```



```
ALTER TABLE ONLY autoservice.car
    ADD CONSTRAINT car_license_plate_key UNIQUE (license_plate);
```

```
--
-- TOC entry 3796 (class 2606 OID 16454)
-- Name: car_model car_model_pkey; Type: CONSTRAINT; Schema:
autoservice; Owner: bareckijmaksim
```

```
--
ALTER TABLE ONLY autoservice.car_model
    ADD CONSTRAINT car_model_pkey PRIMARY KEY (model_id);
```

```
--
-- TOC entry 3802 (class 2606 OID 16464)
-- Name: car car_pkey; Type: CONSTRAINT; Schema: autoservice; Owner:
bareckijmaksim
```

```
--
ALTER TABLE ONLY autoservice.car
    ADD CONSTRAINT car_pkey PRIMARY KEY (car_id);
```

```
--
-- TOC entry 3804 (class 2606 OID 16468)
-- Name: car car_vin_code_key; Type: CONSTRAINT; Schema: autoservice;
Owner: bareckijmaksim
```

```
--
ALTER TABLE ONLY autoservice.car
    ADD CONSTRAINT car_vin_code_key UNIQUE (vin_code);
```

```
--
-- TOC entry 3790 (class 2606 OID 16444)
```

```
-- Name: client client_pkey; Type: CONSTRAINT; Schema: autoservice;  
Owner: bareckijmaksim  
--
```

```
ALTER TABLE ONLY autoservice.client  
    ADD CONSTRAINT client_pkey PRIMARY KEY (client_id);
```

```
--  
-- TOC entry 3817 (class 2606 OID 16552)  
-- Name: contract_line contract_line_pkey; Type: CONSTRAINT; Schema:  
autoservice; Owner: bareckijmaksim  
--
```

```
ALTER TABLE ONLY autoservice.contract_line  
    ADD CONSTRAINT contract_line_pkey PRIMARY KEY (line_id);
```

```
--  
-- TOC entry 3806 (class 2606 OID 16487)  
-- Name: contract contract_pkey; Type: CONSTRAINT; Schema: autoservice;  
Owner: bareckijmaksim  
--
```

```
ALTER TABLE ONLY autoservice.contract  
    ADD CONSTRAINT contract_pkey PRIMARY KEY (contract_id);
```

```
--  
-- TOC entry 3788 (class 2606 OID 16428)  
-- Name: employee_assignment employee_assignment_pkey; Type:  
CONSTRAINT; Schema: autoservice; Owner: bareckijmaksim  
--
```

```
ALTER TABLE ONLY autoservice.employee_assignment  
    ADD CONSTRAINT employee_assignment_pkey PRIMARY KEY  
(assignment_id);
```

--
-- TOC entry 3780 (class 2606 OID 16404)
-- Name: employee_category employee_category_pkey; Type: CONSTRAINT;
Schema: autoservice; Owner: bareckijmaksim
--

ALTER TABLE ONLY autoservice.employee_category
ADD CONSTRAINT employee_category_pkey PRIMARY KEY
(category_id);

--
-- TOC entry 3784 (class 2606 OID 16416)
-- Name: employee employee_personnel_number_key; Type: CONSTRAINT;
Schema: autoservice; Owner: bareckijmaksim
--

ALTER TABLE ONLY autoservice.employee
ADD CONSTRAINT employee_personnel_number_key UNIQUE
(personnel_number);

--
-- TOC entry 3786 (class 2606 OID 16414)
-- Name: employee employee_pkey; Type: CONSTRAINT; Schema:
autoservice; Owner: bareckijmaksim
--

ALTER TABLE ONLY autoservice.employee
ADD CONSTRAINT employee_pkey PRIMARY KEY (employee_id);

--
-- TOC entry 3828 (class 2606 OID 16596)

-- Name: normative_consumption normative_consumption_pkey; Type: CONSTRAINT; Schema: autoservice; Owner: bareckijmaksim

--

ALTER TABLE ONLY autoservice.normative_consumption
ADD CONSTRAINT normative_consumption_pkey PRIMARY KEY
(consumption_id);

--

-- TOC entry 3820 (class 2606 OID 16576)
-- Name: part part_catalog_number_key; Type: CONSTRAINT; Schema: autoservice; Owner: bareckijmaksim

--

ALTER TABLE ONLY autoservice.part
ADD CONSTRAINT part_catalog_number_key UNIQUE (catalog_number);

--

-- TOC entry 3824 (class 2606 OID 16584)
-- Name: part_origin part_origin_part_id_key; Type: CONSTRAINT; Schema: autoservice; Owner: bareckijmaksim

--

ALTER TABLE ONLY autoservice.part_origin
ADD CONSTRAINT part_origin_part_id_key UNIQUE (part_id);

--

-- TOC entry 3826 (class 2606 OID 16582)
-- Name: part_origin part_origin_pkey; Type: CONSTRAINT; Schema: autoservice; Owner: bareckijmaksim

--

ALTER TABLE ONLY autoservice.part_origin
ADD CONSTRAINT part_origin_pkey PRIMARY KEY (origin_id);

--
-- TOC entry 3822 (class 2606 OID 16574)
-- Name: part_part_pkey; Type: CONSTRAINT; Schema: autoservice; Owner:
bareckijmaksim
--

ALTER TABLE ONLY autoservice.part
ADD CONSTRAINT part_pkey PRIMARY KEY (part_id);

--
-- TOC entry 3815 (class 2606 OID 16535)
-- Name: price_period_price_period_pkey; Type: CONSTRAINT; Schema:
autoservice; Owner: bareckijmaksim
--

ALTER TABLE ONLY autoservice.price_period
ADD CONSTRAINT price_period_pkey PRIMARY KEY (period_id);

--
-- TOC entry 3813 (class 2606 OID 16523)
-- Name: service_price_service_price_pkey; Type: CONSTRAINT; Schema:
autoservice; Owner: bareckijmaksim
--

ALTER TABLE ONLY autoservice.service_price
ADD CONSTRAINT service_price_pkey PRIMARY KEY (price_id);

--
-- TOC entry 3809 (class 2606 OID 16513)
-- Name: service_type_service_type_pkey; Type: CONSTRAINT; Schema:
autoservice; Owner: bareckijmaksim
--

```
ALTER TABLE ONLY autoservice.service_type
  ADD CONSTRAINT service_type_pkey PRIMARY KEY (service_type_id);
```

```
--
-- TOC entry 3811 (class 2606 OID 16515)
-- Name: service_type service_type_service_name_key; Type: CONSTRAINT;
Schema: autoservice; Owner: bareckijmaksim
```

```
--
ALTER TABLE ONLY autoservice.service_type
  ADD CONSTRAINT service_type_service_name_key UNIQUE
(service_name);
```

```
--
-- TOC entry 3798 (class 2606 OID 16456)
-- Name: car_model uq_car_model; Type: CONSTRAINT; Schema:
autoservice; Owner: bareckijmaksim
```

```
--
ALTER TABLE ONLY autoservice.car_model
  ADD CONSTRAINT uq_car_model UNIQUE (brand, model_name,
manufacturer);
```

```
--
-- TOC entry 3792 (class 2606 OID 16448)
-- Name: client uq_client_email; Type: CONSTRAINT; Schema: autoservice;
Owner: bareckijmaksim
```

```
--
ALTER TABLE ONLY autoservice.client
  ADD CONSTRAINT uq_client_email UNIQUE (email);
```

--
-- TOC entry 3794 (class 2606 OID 16446)
-- Name: client uq_client_phone; Type: CONSTRAINT; Schema: autoservice;
Owner: bareckijmaksim
--

ALTER TABLE ONLY autoservice.client
ADD CONSTRAINT uq_client_phone UNIQUE (phone);

--
-- TOC entry 3782 (class 2606 OID 16406)
-- Name: employee_category uq_employee_category; Type: CONSTRAINT;
Schema: autoservice; Owner: bareckijmaksim
--

ALTER TABLE ONLY autoservice.employee_category
ADD CONSTRAINT uq_employee_category UNIQUE (position_name,
specialization, grade);

--
-- TOC entry 3830 (class 2606 OID 16598)
-- Name: normative_consumption uq_normative; Type: CONSTRAINT;
Schema: autoservice; Owner: bareckijmaksim
--

ALTER TABLE ONLY autoservice.normative_consumption
ADD CONSTRAINT uq_normative UNIQUE (service_type_id, part_id);

--
-- TOC entry 3832 (class 2606 OID 16617)
-- Name: used_part used_part_pkey; Type: CONSTRAINT; Schema:
autoservice; Owner: bareckijmaksim
--

```
ALTER TABLE ONLY autoservice.used_part
  ADD CONSTRAINT used_part_pkey PRIMARY KEY (used_part_id);
```

```
--
-- TOC entry 3776 (class 2606 OID 16398)
-- Name: workshop workshop_email_key; Type: CONSTRAINT; Schema:
autoservice; Owner: bareckijmaksim
--
```

```
ALTER TABLE ONLY autoservice.workshop
  ADD CONSTRAINT workshop_email_key UNIQUE (email);
```

```
--
-- TOC entry 3778 (class 2606 OID 16396)
-- Name: workshop workshop_pkey; Type: CONSTRAINT; Schema:
autoservice; Owner: bareckijmaksim
--
```

```
ALTER TABLE ONLY autoservice.workshop
  ADD CONSTRAINT workshop_pkey PRIMARY KEY (workshop_id);
```

```
--
-- TOC entry 3818 (class 1259 OID 24588)
-- Name: idx_contract_line_contract_service; Type: INDEX; Schema:
autoservice; Owner: bareckijmaksim
--
```

```
CREATE INDEX idx_contract_line_contract_service ON
autoservice.contract_line USING btree (contract_id, service_type_id);
```

```
--
-- TOC entry 3807 (class 1259 OID 24587)
```


-- Name: idx_contract_status_date; Type: INDEX; Schema: autoservice; Owner: bareckijmaksim

--

CREATE INDEX idx_contract_status_date ON autoservice.contract USING btree (contract_status, acceptance_date);

--

-- TOC entry 3834 (class 2606 OID 16434)

-- Name: employee_assignment fk_assignment_category; Type: FK CONSTRAINT; Schema: autoservice; Owner: bareckijmaksim

--

ALTER TABLE ONLY autoservice.employee_assignment
ADD CONSTRAINT fk_assignment_category FOREIGN KEY
(category_id) REFERENCES autoservice.employee_category(category_id) ON
UPDATE CASCADE ON DELETE RESTRICT;

--

-- TOC entry 3835 (class 2606 OID 16429)

-- Name: employee_assignment fk_assignment_employee; Type: FK CONSTRAINT; Schema: autoservice; Owner: bareckijmaksim

--

ALTER TABLE ONLY autoservice.employee_assignment
ADD CONSTRAINT fk_assignment_employee FOREIGN KEY
(employee_id) REFERENCES autoservice.employee(employee_id) ON
UPDATE CASCADE ON DELETE CASCADE;

--

-- TOC entry 3836 (class 2606 OID 16469)

-- Name: car fk_car_client; Type: FK CONSTRAINT; Schema: autoservice; Owner: bareckijmaksim

--

```
ALTER TABLE ONLY autoservice.car
    ADD CONSTRAINT fk_car_client FOREIGN KEY (client_id)
REFERENCES autoservice.client(client_id) ON UPDATE CASCADE ON
DELETE RESTRICT;
```

```
--
-- TOC entry 3837 (class 2606 OID 16474)
-- Name: car fk_car_model; Type: FK CONSTRAINT; Schema: autoservice;
Owner: bareckijmaksim
--
```

```
ALTER TABLE ONLY autoservice.car
    ADD CONSTRAINT fk_car_model FOREIGN KEY (model_id)
REFERENCES autoservice.car_model(model_id) ON UPDATE CASCADE
ON DELETE RESTRICT;
```

```
--
-- TOC entry 3838 (class 2606 OID 16503)
-- Name: contract fk_contract_admin; Type: FK CONSTRAINT; Schema:
autoservice; Owner: bareckijmaksim
--
```

```
ALTER TABLE ONLY autoservice.contract
    ADD CONSTRAINT fk_contract_admin FOREIGN KEY
(administrator_employee_id) REFERENCES
autoservice.employee(employee_id) ON UPDATE CASCADE ON DELETE
RESTRICT;
```

```
--
-- TOC entry 3839 (class 2606 OID 16498)
-- Name: contract fk_contract_car; Type: FK CONSTRAINT; Schema:
autoservice; Owner: bareckijmaksim
--
```

```
ALTER TABLE ONLY autoservice.contract
  ADD CONSTRAINT fk_contract_car FOREIGN KEY (car_id)
REFERENCES autoservice.car(car_id) ON UPDATE CASCADE ON DELETE
RESTRICT;
```

```
--
-- TOC entry 3840 (class 2606 OID 16493)
-- Name: contract fk_contract_client; Type: FK CONSTRAINT; Schema:
autoservice; Owner: bareckijmaksim
--
```

```
ALTER TABLE ONLY autoservice.contract
  ADD CONSTRAINT fk_contract_client FOREIGN KEY (client_id)
REFERENCES autoservice.client(client_id) ON UPDATE CASCADE ON
DELETE RESTRICT;
```

```
--
-- TOC entry 3845 (class 2606 OID 16553)
-- Name: contract_line fk_contract_line_contract; Type: FK CONSTRAINT;
Schema: autoservice; Owner: bareckijmaksim
--
```

```
ALTER TABLE ONLY autoservice.contract_line
  ADD CONSTRAINT fk_contract_line_contract FOREIGN KEY
(contract_id) REFERENCES autoservice.contract(contract_id) ON UPDATE
CASCADE ON DELETE CASCADE;
```

```
--
-- TOC entry 3846 (class 2606 OID 16558)
-- Name: contract_line fk_contract_line_employee; Type: FK CONSTRAINT;
Schema: autoservice; Owner: bareckijmaksim
--
```

```
ALTER TABLE ONLY autoservice.contract_line
  ADD CONSTRAINT fk_contract_line_employee FOREIGN KEY
(employee_id) REFERENCES autoservice.employee(employee_id) ON
UPDATE CASCADE ON DELETE RESTRICT;
```

```
--
-- TOC entry 3847 (class 2606 OID 16563)
-- Name: contract_line fk_contract_line_service; Type: FK CONSTRAINT;
Schema: autoservice; Owner: bareckijmaksim
```

```
ALTER TABLE ONLY autoservice.contract_line
  ADD CONSTRAINT fk_contract_line_service FOREIGN KEY
(service_type_id) REFERENCES autoservice.service_type(service_type_id)
ON UPDATE CASCADE ON DELETE RESTRICT;
```

```
--
-- TOC entry 3841 (class 2606 OID 16488)
-- Name: contract fk_contract_workshop; Type: FK CONSTRAINT; Schema:
autoservice; Owner: bareckijmaksim
```

```
ALTER TABLE ONLY autoservice.contract
  ADD CONSTRAINT fk_contract_workshop FOREIGN KEY (workshop_id)
REFERENCES autoservice.workshop(workshop_id) ON UPDATE CASCADE
ON DELETE RESTRICT;
```

```
--
-- TOC entry 3833 (class 2606 OID 16417)
-- Name: employee fk_employee_workshop; Type: FK CONSTRAINT;
Schema: autoservice; Owner: bareckijmaksim
```

```
ALTER TABLE ONLY autoservice.employee
```

```
ADD CONSTRAINT fk_employee_workshop FOREIGN KEY
(workshop_id) REFERENCES autoservice.workshop(workshop_id) ON
UPDATE CASCADE ON DELETE RESTRICT;
```

--

-- TOC entry 3849 (class 2606 OID 16604)

-- Name: normative_consumption fk_normative_part; Type: FK
CONSTRAINT; Schema: autoservice; Owner: bareckijmaksim

--

```
ALTER TABLE ONLY autoservice.normative_consumption
```

```
ADD CONSTRAINT fk_normative_part FOREIGN KEY (part_id)
REFERENCES autoservice.part(part_id) ON UPDATE CASCADE ON
DELETE RESTRICT;
```

--

-- TOC entry 3850 (class 2606 OID 16599)

-- Name: normative_consumption fk_normative_service; Type: FK
CONSTRAINT; Schema: autoservice; Owner: bareckijmaksim

--

```
ALTER TABLE ONLY autoservice.normative_consumption
```

```
ADD CONSTRAINT fk_normative_service FOREIGN KEY
(service_type_id) REFERENCES autoservice.service_type(service_type_id)
ON UPDATE CASCADE ON DELETE CASCADE;
```

--

-- TOC entry 3848 (class 2606 OID 16585)

-- Name: part_origin fk_part_origin_part; Type: FK CONSTRAINT; Schema:
autoservice; Owner: bareckijmaksim

--

```
ALTER TABLE ONLY autoservice.part_origin
```

```
ADD CONSTRAINT fk_part_origin_part FOREIGN KEY (part_id)
REFERENCES autoservice.part(part_id) ON UPDATE CASCADE ON
DELETE CASCADE;
```

```
--
```

```
-- TOC entry 3843 (class 2606 OID 16541)
```

```
-- Name: price_period fk_price_period_price; Type: FK CONSTRAINT;
Schema: autoservice; Owner: bareckijmaksim
```

```
--
```

```
ALTER TABLE ONLY autoservice.price_period
```

```
ADD CONSTRAINT fk_price_period_price FOREIGN KEY (price_id)
REFERENCES autoservice.service_price(price_id) ON UPDATE CASCADE
ON DELETE CASCADE;
```

```
--
```

```
-- TOC entry 3844 (class 2606 OID 16536)
```

```
-- Name: price_period fk_price_period_service; Type: FK CONSTRAINT;
Schema: autoservice; Owner: bareckijmaksim
```

```
--
```

```
ALTER TABLE ONLY autoservice.price_period
```

```
ADD CONSTRAINT fk_price_period_service FOREIGN KEY
(service_type_id) REFERENCES autoservice.service_type(service_type_id)
ON UPDATE CASCADE ON DELETE CASCADE;
```

```
--
```

```
-- TOC entry 3842 (class 2606 OID 16524)
```

```
-- Name: service_price fk_service_price_service; Type: FK CONSTRAINT;
Schema: autoservice; Owner: bareckijmaksim
```

```
--
```

```
ALTER TABLE ONLY autoservice.service_price
```

```
    ADD CONSTRAINT fk_service_price_service FOREIGN KEY
(service_type_id) REFERENCES autoservice.service_type(service_type_id)
ON UPDATE CASCADE ON DELETE CASCADE;
```

```
--
```

```
-- TOC entry 3851 (class 2606 OID 16618)
```

```
-- Name: used_part fk_used_part_line; Type: FK CONSTRAINT; Schema:
autoservice; Owner: bareckijmaksim
```

```
--
```

```
ALTER TABLE ONLY autoservice.used_part
```

```
    ADD CONSTRAINT fk_used_part_line FOREIGN KEY (line_id)
REFERENCES autoservice.contract_line(line_id) ON UPDATE CASCADE
ON DELETE CASCADE;
```

```
--
```

```
-- TOC entry 3852 (class 2606 OID 16623)
```

```
-- Name: used_part fk_used_part_part; Type: FK CONSTRAINT; Schema:
autoservice; Owner: bareckijmaksim
```

```
--
```

```
ALTER TABLE ONLY autoservice.used_part
```

```
    ADD CONSTRAINT fk_used_part_part FOREIGN KEY (part_id)
REFERENCES autoservice.part(part_id) ON UPDATE CASCADE ON
DELETE RESTRICT;
```

```
-- Completed on 2026-04-13 17:19:07 MSK
```

```
--
```

```
-- PostgreSQL database dump complete
```

```
--
```

\unrestrict

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